Soundarray Tech Report

Concept-first architecture for spatial audio + classification

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Abstract. soundarray explores spatial audio processing and sound classification with an emphasis on edge capture (Raspberry Pi + microphone arrays) and optional remote compute. The design goal is inspectable, reproducible inference: signals are windowed explicitly, intermediate representations are named, and each stage has testable invariants.

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1 Problem Statement

We want a system that can (a) capture multi-channel audio from an array, (b) infer spatial properties (e.g., direction-of-arrival), (c) denoise/enhance, and (d) classify sources (vehicles, wildlife, etc.) under real-world constraints.

2 System Model (Concept-First)

Insight:
Prefer diagrams of predicates, invariants, and consistency boundaries. Avoid UML class diagrams and linear flowcharts: they hide the actual correctness contract.

The system is best understood as a set of signal transformations plus a small number of serialization points (places where we intentionally collapse concurrency into a stable, deterministic order).

2.1 End-to-End Dataflow: Predicates and Signal Pipelines

Use this as the mental model: a *predicate* ("there is a source of type X at bearing θ") is derived from multiple windowed signals and gates.

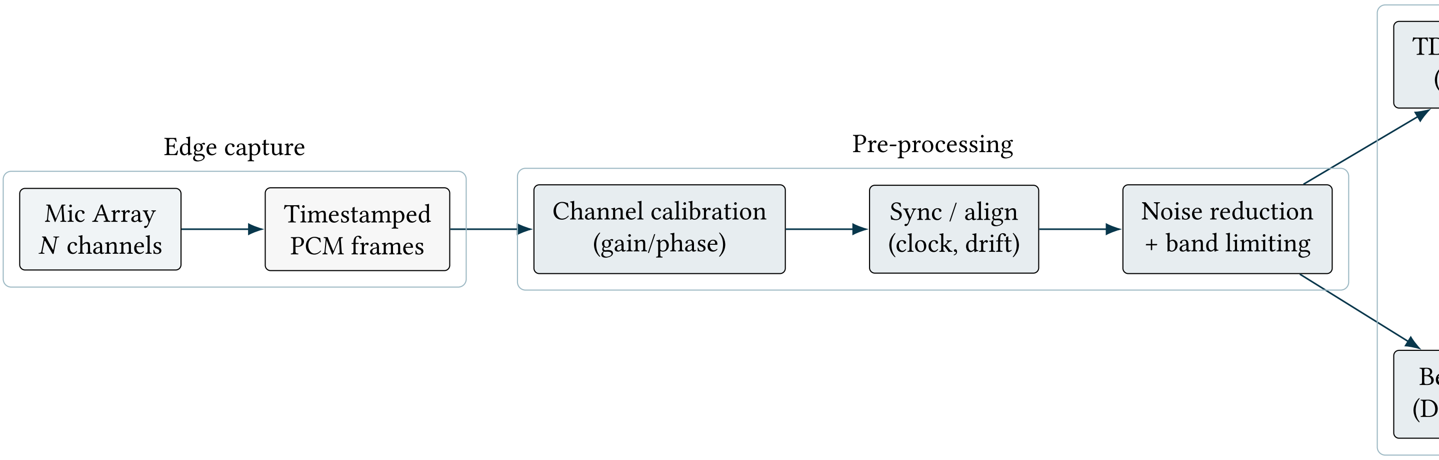


Figure 1. Predicate-first pipeline. The output is a provenance-bearing predicate over time (events), not "a label".

2.2 Consistency Boundaries: Snapshot-Then-Apply

Spatial inference is sensitive to window boundaries. We treat each analysis tick as: (1) snapshot a well-defined window $W = [t_0, t_1)$, then (2) apply derived updates atomically.

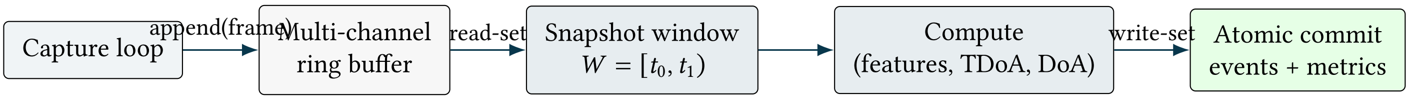


Figure 2. A single serialization point: snapshot a coherent window, then commit derived outputs atomically for tick k.

3 Core Invariants (What Must Always Hold)

: Window coherence

For any analysis window W, all channels represent the same real-world time interval. Resampling or drift correction is applied consistently across channels.

: Stable framing

Frames have explicit sample rate, channel count, and a monotonic time anchor. Dropped frames are detected and recorded as gaps (never silently ignored).

: Geometry provenance

Array geometry and per-mic calibration parameters are versioned and included in the analysis configuration referenced by outputs.

: Reproducible ticks

Given identical PCM bytes and identical configuration, derived features and event outputs are identical (within a documented floating-point tolerance).

4 Algorithmic Spine (Advanced Ideas)

4.1 TDoA via GCC-PHAT (Why It Works)

Goal: estimate a time delay Δt between two microphone signals robustly under reverberation.

Concept model:

- compute cross-power spectrum $X_{12}(\omega)$
- apply PHAT weighting to emphasize phase alignment
- inverse FFT to obtain a correlation peak; $\arg\max$ gives Δt

Correctness depends on window coherence and sensible band limiting (avoid bands that produce spurious peaks).

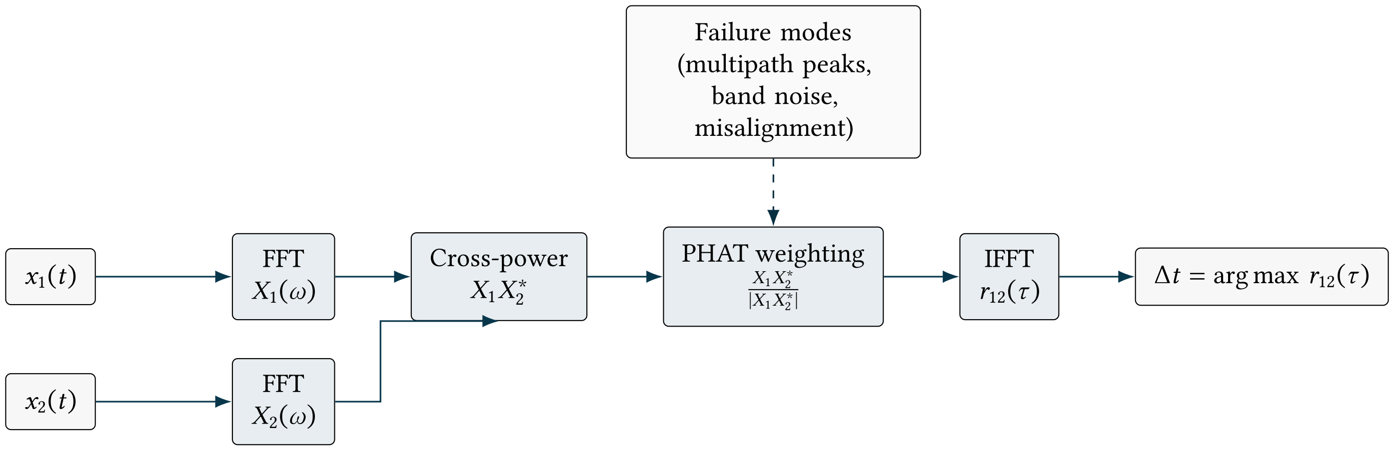


Figure 3. GCC-PHAT as a transform stack. This diagram communicates the semantics: we estimate a delay by constructing a robust correlation peak, not by matching amplitudes.

4.2 Beamforming as Rewrite Semantics

Rather than "run a beamformer", view beamforming as a deterministic rewrite:

- input: multi-channel frame + steering for angle θ
- rewrite: phase-align channels according to expected propagation delays
- output: an enhanced signal $y(t;\theta)$

This frames the correctness contract: geometry + timing + steering must agree.

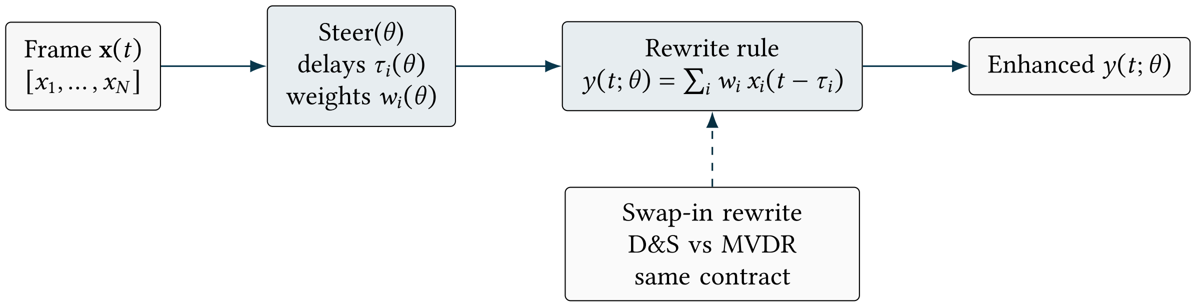


Figure 4. Beamforming as deterministic rewrite semantics. Implementations differ, but the contract is invariant: geometry + timing + steering define the transform.

4.3 Classification as a Gated Predicate

Classification is one gate among several quality/stability gates:

- SNR/quality metrics must exceed thresholds
- spatial estimate must be stable over a debounce window
- classifier probabilities must exceed class-specific thresholds

The output is not "a label"; it is a predicate with provenance.

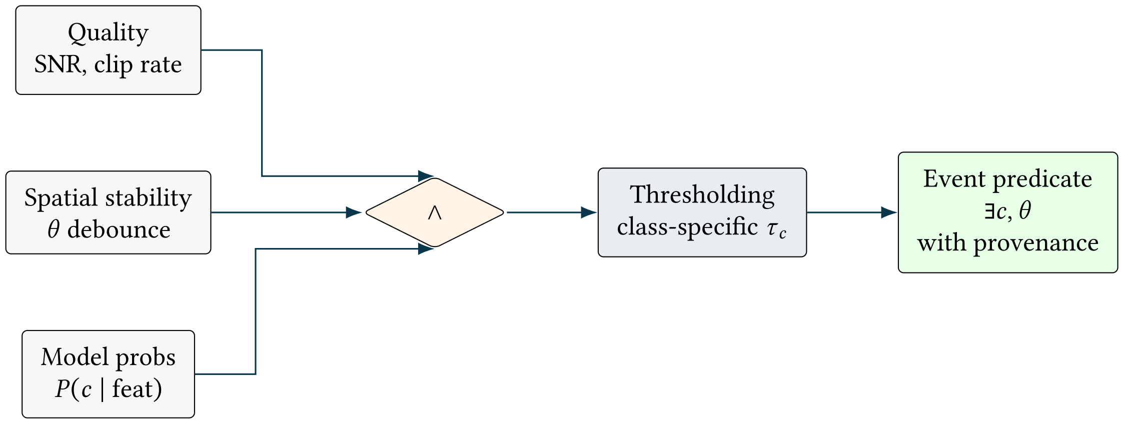


Figure 5. Classification as a gated predicate. The system emits events only when multiple correctness conditions hold over the same window.

4.4 Array Geometry and Delay Manifold

Spatial algorithms rest on geometry. The key derived object is the delay manifold $\tau_i(\theta)$ induced by a plane wave at bearing θ.

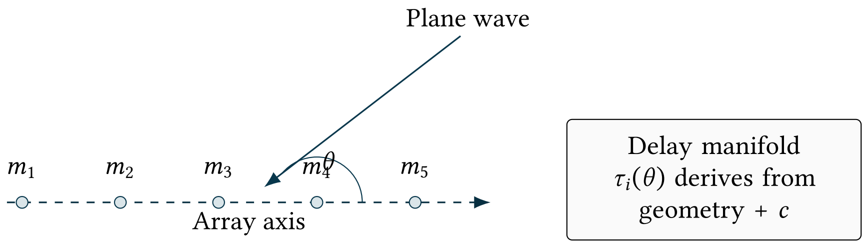


Figure 6. Array geometry induces a delay manifold. Spatial inference is meaningful only if geometry provenance and time alignment invariants hold.

4.5 Consistency Boundary as Happens-Before Timeline

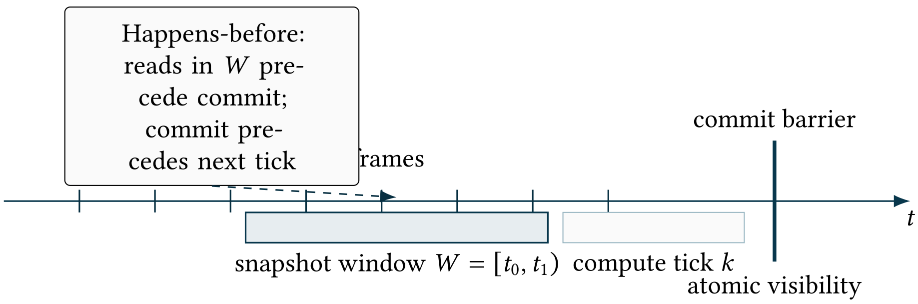


Figure 7. Consistency boundary as a happens-before model. The snapshot and commit define the determinism anchor for replay.

4.6 Uncertainty Propagation / Error Budget

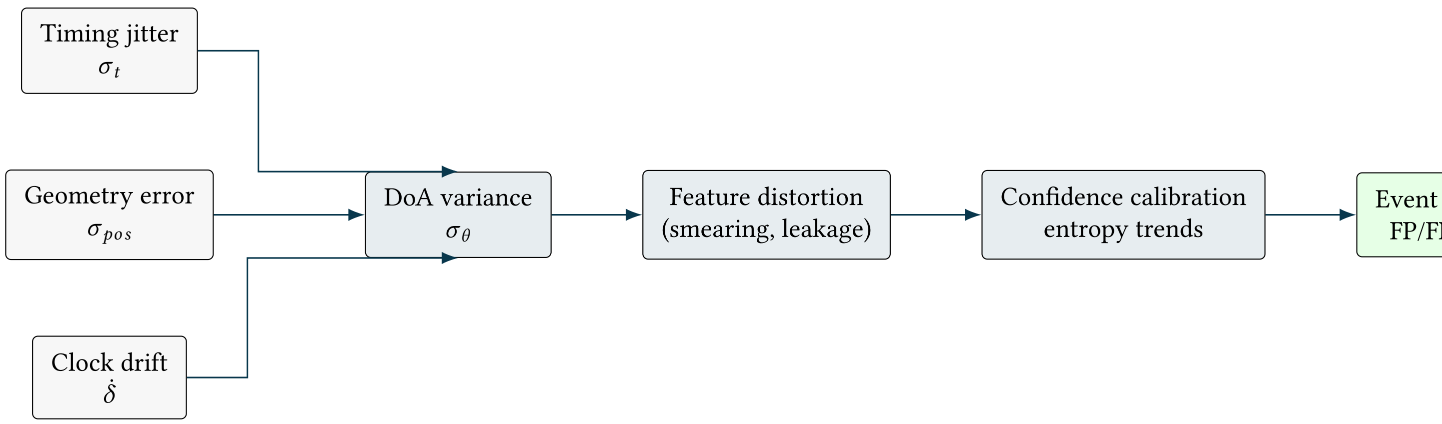


Figure 8. Error budget as propagation. Downstream certainty is bounded by upstream timing and geometry uncertainty; measure these explicitly.

4.7 Replay and Traceability DAG

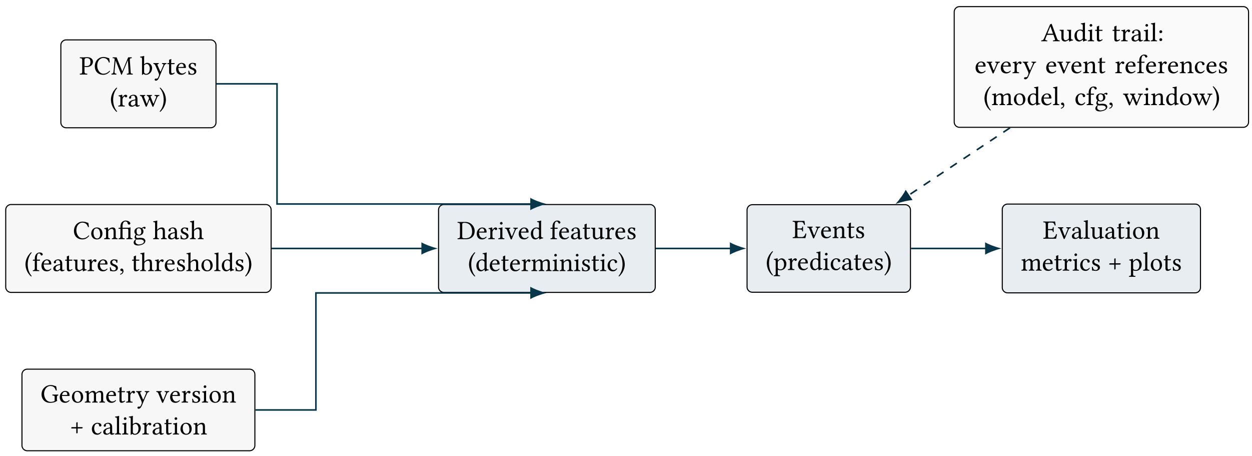


Figure 9. Replay and traceability as a DAG. Deterministic transforms plus versioned inputs make results reproducible and debuggable.

5 Deployment Shapes (Edge vs Remote)

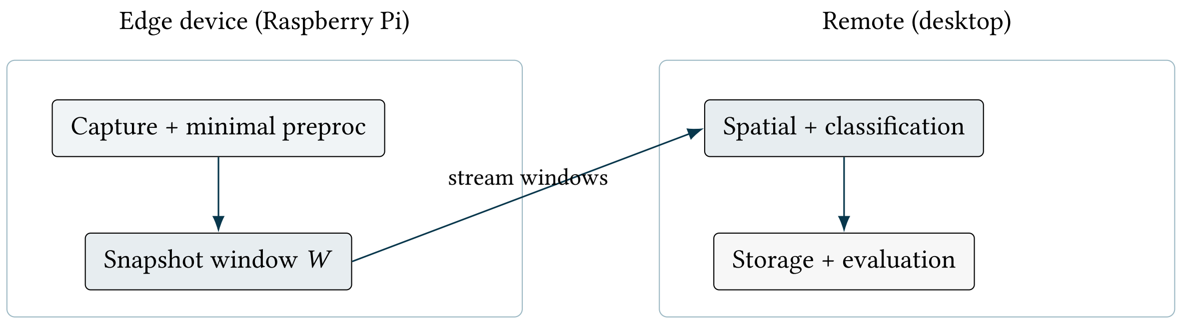


Figure 10. Two deployment shapes. Prefer placing the determinism anchor (window snapshot) as close to capture as possible.

6 Observability (What To Measure)

Treat metrics as first-class outputs:

- capture: frame drops, jitter, drift estimates
- preproc: per-channel RMS, noise floor, clipping rate
- spatial: GCC-PHAT peak sharpness, bearing stability, beamformer gain
- classifier: confidence/entropy trends, false trigger rate

7 Open Questions / Next Steps

- define a replay interchange format (multichannel WAV + JSON sidecar for geometry, framing, timestamps)
- choose and freeze canonical feature transforms (window, hop, mel bins, normalization) for determinism
- add an evaluation harness: known recordings + expected event traces

8 Additional Concept Diagrams (Failure Modes and Boundaries)

This section adds a few diagrams that are especially useful in practice: they surface failure modes, regime boundaries, and the engineering gates that preserve correctness.

8.1 Multipath Ambiguity (Why Peaks Multiply)

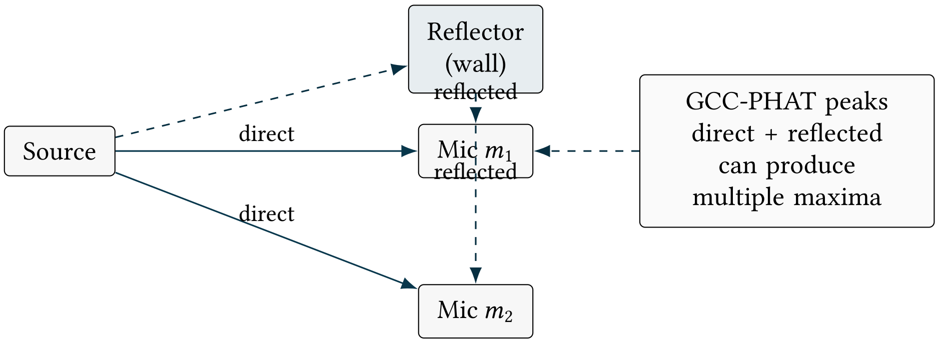


Figure 11. Multipath creates multiple plausible delays. Peak selection should be constrained by physically plausible bounds derived from array aperture and speed of sound.

8.2 Near-Field vs Far-Field Boundary

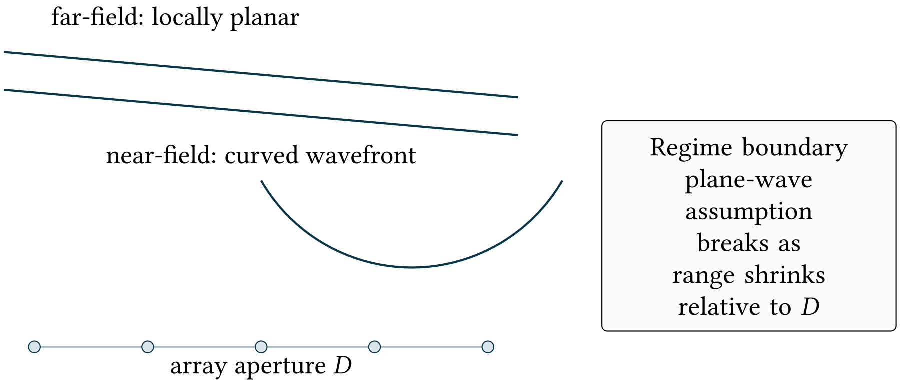


Figure 12. Near-field vs far-field is a model boundary. If the plane-wave assumption fails, DoA-only inference may be biased and a range-bearing model becomes appropriate.

8.3 Beam Pattern Sketch (Selectivity vs Robustness)

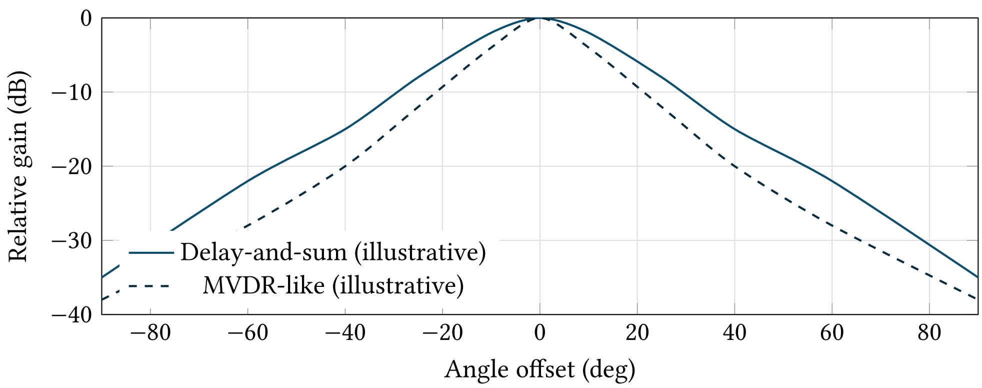


Figure 13. Beam pattern intuition: stronger selectivity suppresses off-axis energy but may be less robust to calibration/geometry error. In real evaluation, compute these curves from the actual array and steering model.

8.4 STFT Windowing Semantics (Leakage vs Resolution)

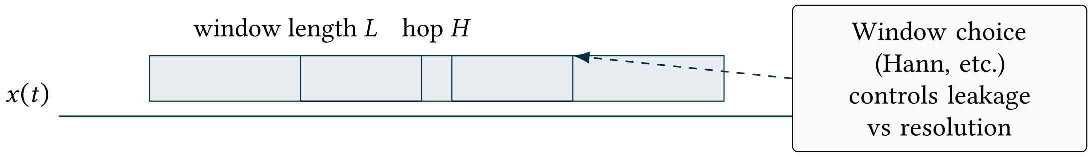


Figure 14. STFT semantics: features are defined by window length L, hop H, and window function. These parameters are part of the determinism contract.

8.5 Calibration Loop as an Invariant-Preserving Update

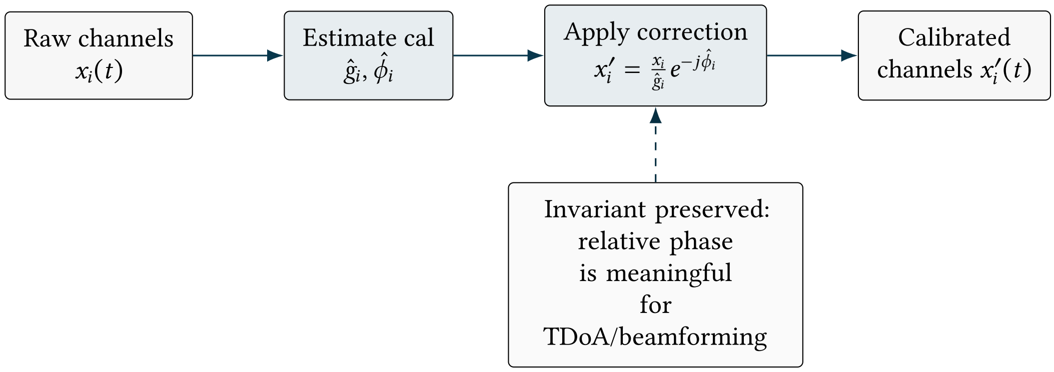


Figure 15. Calibration is not "preprocessing"; it is an invariant-preserving update that makes cross-channel phase comparisons meaningful.

8.6 Edge-to-Remote Streaming Boundary (Window Assembly)

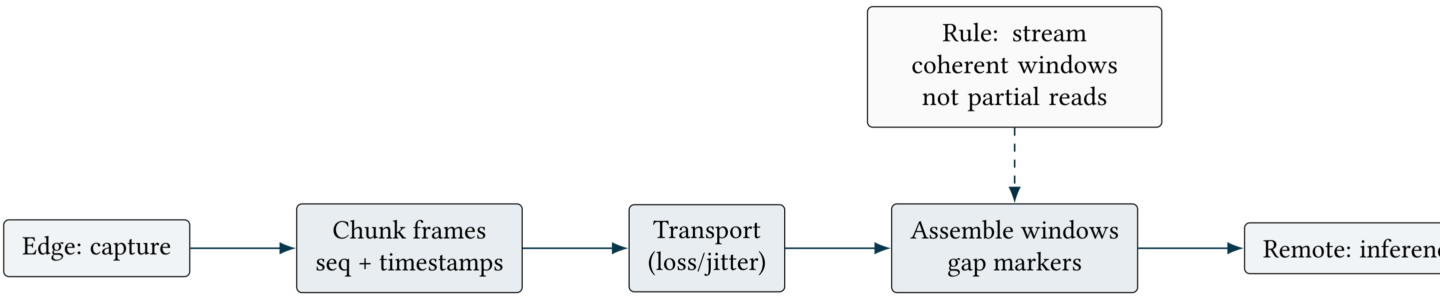


Figure 16. Streaming boundary diagram. To preserve window coherence, remote inference must operate on assembled windows with explicit gap semantics.

8.7 Evaluation Harness as a Workflow Gate

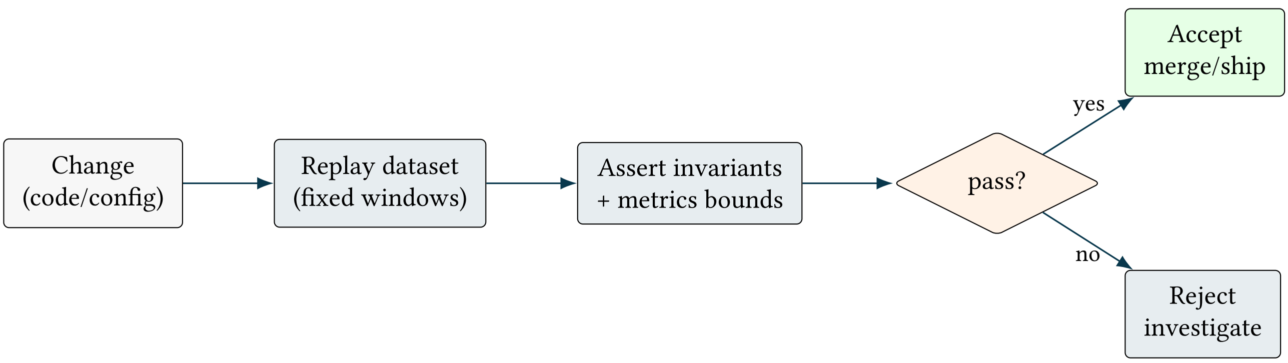


Figure 17. Workflow as gates. This makes correctness operational: changes are accepted only if replay preserves invariants and stays within known-good metric envelopes.

8.8 DoA Identifiability (Symmetry and Spatial Aliasing)

Some DoA failures are not bugs; they are non-identifiability induced by geometry or frequency.

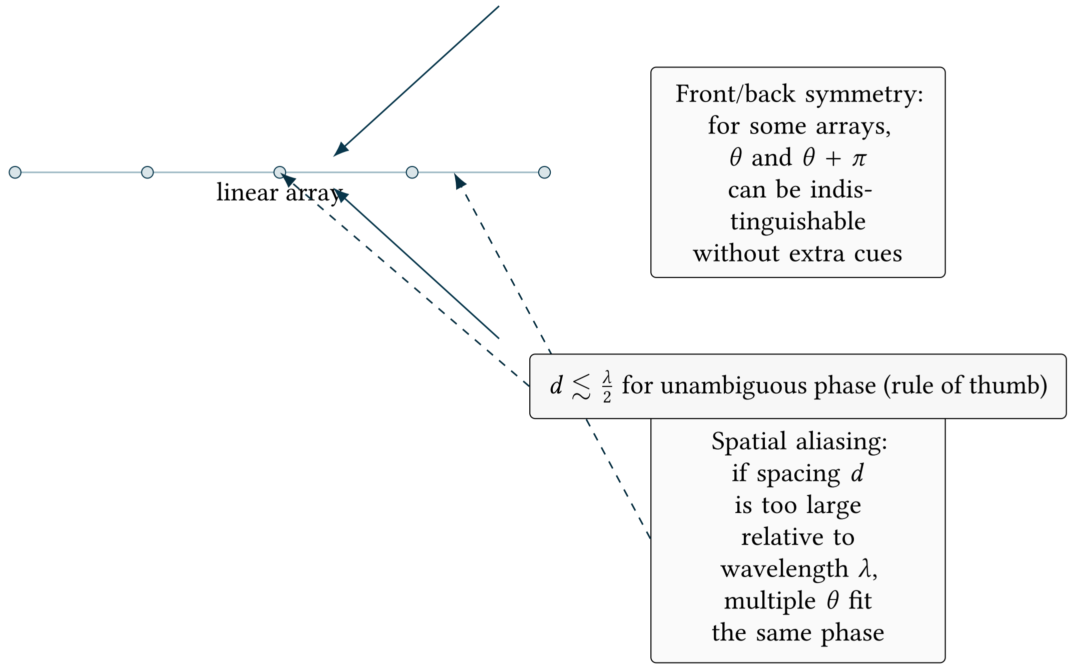


Figure 18. DoA identifiability constraints. Symmetry can cause 180-degree ambiguity; large inter-mic spacing can cause spatial aliasing at higher frequencies. Use geometry choices and auxiliary cues (asymmetry, higher-order arrays, or priors) to break ambiguity.

8.9 Speed-of-Sound Sensitivity (Environment → Bias)

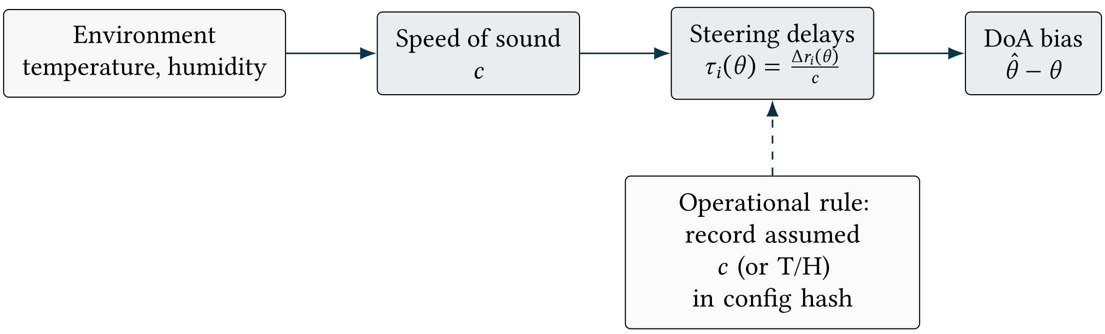


Figure 19. Environment changes propagate into steering delays through c. If c is implicitly assumed, DoA and beamforming can drift in a way that looks like a model regression. Treat c as a versioned configuration input.

8.10 Coherence as a Precondition (When Spatial Inference Is Meaningful)

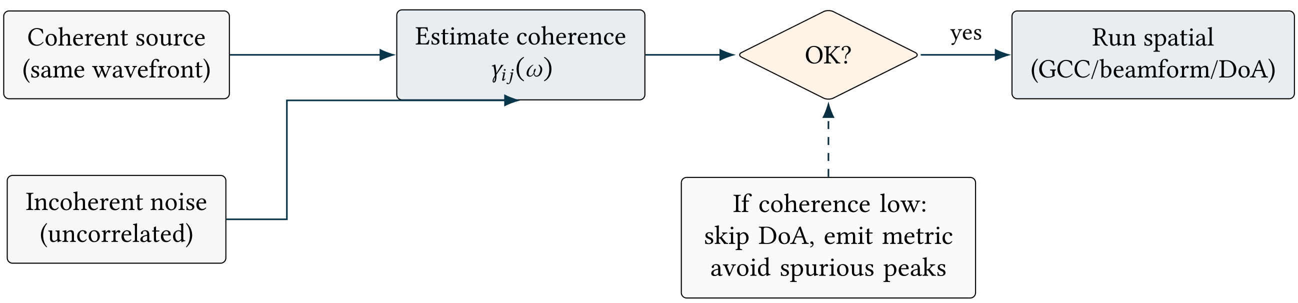


Figure 20. Coherence is a precondition: if signals are dominated by incoherent noise, spatial inference is ill-posed. Treat coherence as a gate that prevents false DoA triggers.

8.11 Beamforming vs Source Separation (Boundary of Expectations)

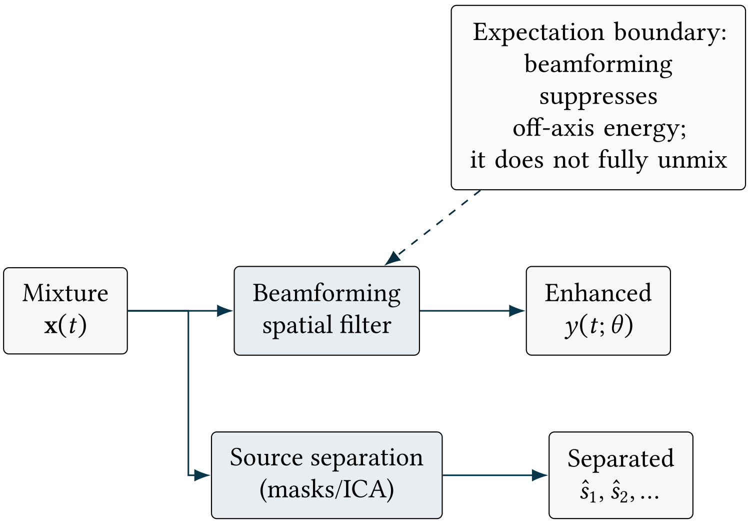


Figure 21. Beamforming and source separation solve different problems. Conflating them causes unrealistic expectations and misinterpreted failures.

8.12 Latency Budget Decomposition (Where Time Goes)

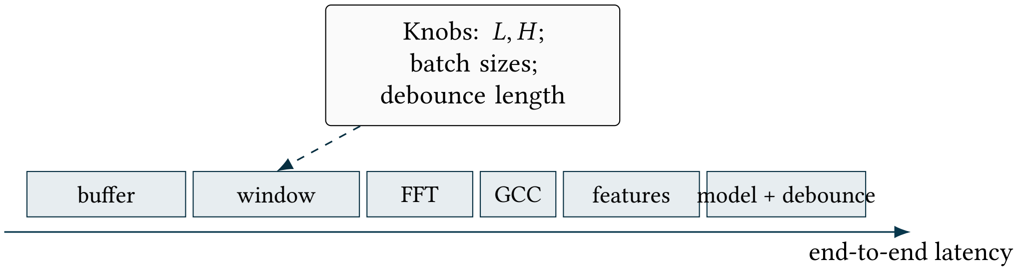


Figure 22. Latency budget. Window length and debounce are often the dominant terms; treat them as explicit tradeoffs against accuracy/stability.

8.13 Configuration as a First-Class Artifact (Hash → Provenance)

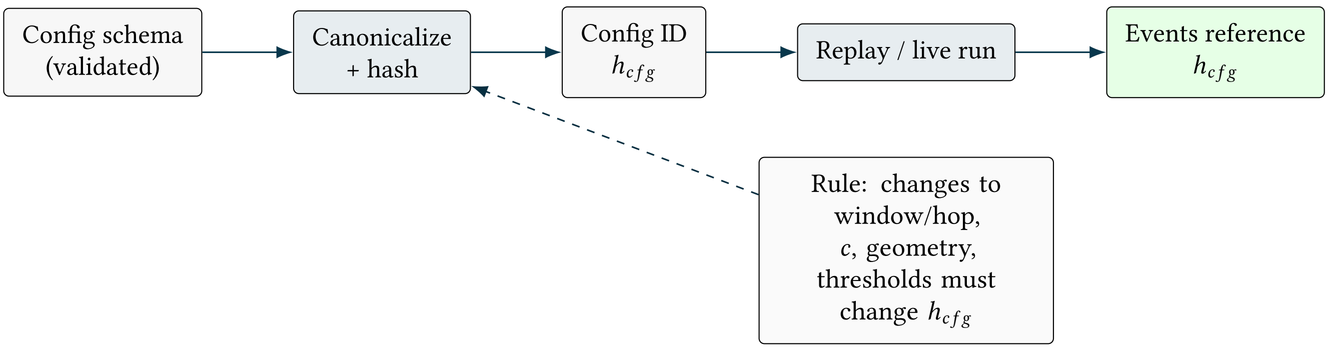


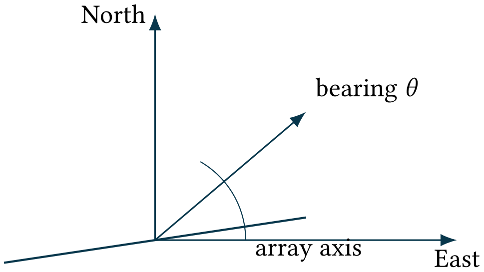
Figure 23. Configuration as an artifact. Provenance is operational only if configuration is canonicalized and hashed, and every output references the hash.

8.14 Failure Taxonomy Matrix (Invariants × Symptoms)

Invariant broken	Common symptom	Secondary symptom	What to measure first
Window coherence	DoA jitter / unstable peaks	classifier confidence oscillates	drop rate, drift estimate
Geometry provenance	consistent bias in θ	beamforming gain lower than expected	geometry version, orientation
Calibration	180 flips / multi-peaks	MVDR unstable	per-channel phase / gain
Determinism	replay mismatch	non-repro metrics	config hash, window IDs

Figure 24. Failure taxonomy. This converts debugging into a lookup: symptom → broken invariant → first measurement.

8.15 Ground-Truth Coordinate Frame (What Does Bearing Mean?)



Define conventions:
0 deg = East? North?
clockwise positive?
elevation ignored/handled?

Figure 25. Coordinate frame definition prevents evaluation mismatches. Bearing semantics must be explicit for ground truth, outputs, and plots.